# Optical Isomerism

1. CHIRAL = Symmetry, need not have chiral centre

2. Optical Isomers → Different activity towards PPL (Both enantiomers + Meso)

3. $+\theta$ ⇒ Right rein rotation ⇒ dextrorotary (d)
   $-\theta$ ⇒ Left rein rotation ⇒ laevorotarory (l)

4. Specific Rotation → $\alpha = \frac{\theta}{l \times c}$ ← observed rotation
   (l: length of tube, c: conc°)

5. STEREOISOMERS

| Enantiomers | Diastereomers |
|---|---|
| Non-superimposable mirror images | No mirror image relation. |

6. Priority → a. At. no
   b. Isotopes → mass no.
   c. Double/Triple to be split into single bonds.
   Use phantom atom ($Z = 0$) to complete the valancy.

eg.- ① $-\overset{\overset{O}{\|}}{C}-OH \Rightarrow -\overset{\overset{O-C_{000}}{|}}{\underset{\underset{O_0}{|}}{C}}-OH$ ② $-C\equiv CH \Rightarrow -\overset{\overset{C_0}{|}}{\underset{\underset{C_{00}}{|}}{C}}-\overset{\overset{C_{000}}{|}}{\underset{\underset{C_{00}}{|}}{C}}-H$ ('0' is zero.)

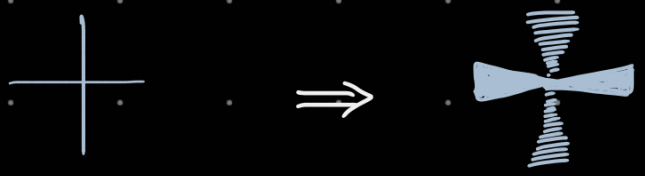
OMP PET → T. Butyl ⇒

ortho, meta, para → tolyl; Ethynyl; Phenyl




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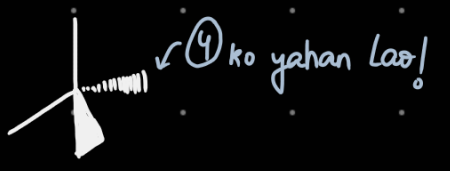
## 7. Fischer Projection : "Arms coming out to hug you!



How to convert?
↳ R/S ✓

## 8. R-S Nomenclature

a. Perspective formula → 1. 4th priority waale ko peeche (Hatch) bhejo
2. Priority (1→2→3) ghumado.




- Clockwise → Cancer → R
- Anti-clockwise → AIDS → S

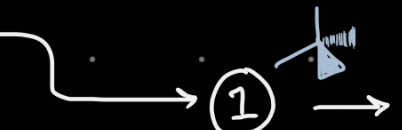
b. Fischer projection → ④ ko vertical pe rkkho! If horizontal ↳ ulta karo!

(1→2→3) → R/S



(1→2→3) → R/S —ulta→ S/R

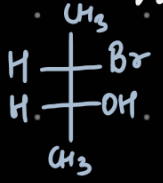
## 9. No. of chiral carbon




2

**Groups different hain ⇒ 4 Opt. isomers (all 4 active)**

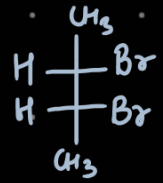
eg:




Diastereomers
- [RR, SS] enantiomers
- [RS, SR] enantiomers

**Groups same hain! ⇒ 3 iso. (2 active 1 meso)**

eg.



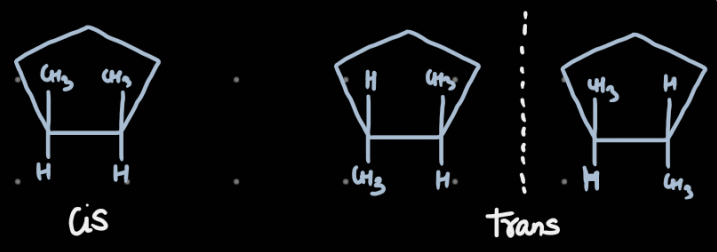

- [RR, SS] enantiomers
- [RS, SR] same! → meso

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10. Geometrical + optical

→ Cis/Trans banao ⇒ Check karo agar O.A ho toh.

eg.




Cis Trans

11. Erythro and threo

↳ Same group Same side

↳ Same group, differ side




12. Allenes and cycloalkane

↳ even no. of π Bonds ⇒ O.A ⇒ not in the same plane




* odd no. ⇒ O.I but Geometrical Iso. ✓✓

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